Battery system installation, storage and transportation manual (Volume 1)

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Ⅰ、Preface

Thank you for choosing the battery system provided by REPT BATTERO Energy Co., Ltd .we will wholeheartedly provide you with high-quality products and services.

- Dear customer, please read the manual carefully before using this product.
- Dear customer, please carefully check and confirm the following when you receive the battery system:
    a. Whether the outer box is intact;
    b. Whether the appearance of the product is intact;
    c. Whether the packing list and the accessories in the box are complete, including product instruction manuals, factory inspection reports, etc.;
    d. For packaging materials and other parts that are marked or declared in advance as needing to be recycled, please keep them properly for us before we recycle, if they are damaged or lost by more than a certain percentage, a certain fee will be charged;
    e. For packaging materials that do not need to be recycled, it is also recommended to keep them for a certain period of time for use during transportation or repairs
- Dear customer, when charging the battery system, please use the charging equipment approved by both parties.
- If the battery system is not used for a long time, turn off the power to the system, disconnect the load, and store the battery system under 25%-40% SOC in a room with an ambient temperature of 25±5°C; It is recommended to check the battery system once a month to ensure that the cell voltage is within the range of 3.2±0.1V, and if the voltage of any cell is lower than 2.8V, it needs to be charged immediately. （Measure：REPT provides a host computer to supply power and read the battery voltage.）
- Please read and abide by the contents of the symbols  in this manual.

II、Safe Use Instructions

- It should be ensured that the working environment of the battery system meets the requirements in the product specification and shall not be used in abnormal scenes (such as high altitude （>3000m）, high salt spray （coastal areas）, high pollution(areas beyond IP67 protection), high humidity(RH>95%), high temperature(>55℃), low temperature(<-20℃), etc.); If there are the above application scenarios, you need to inform our company in advance, and our relevant technical personnel will respond after evaluation.
- The battery system should be kept away from heat and fire sources and avoid direct sunlight for a long time.
- Do not decompose, puncture, crush, roll, incinerate, throw into water or wet the battery.
- Do not hit, throw or cause the battery to be subjected to mechanical vibration and natural fall, to avoid mechanical damage from the outside of the battery box or bumps that cause damage to the anti-corrosion layer of the shell.
- If the battery emits peculiar odor, heat, deformation, discoloration or any other abnormal behavior, stop using it immediately and transfer the battery out of the use environment.
- It is forbidden to connect the positive and negative poles of the battery box or battery system directly with metal or other conductors to avoid ignition or short circuit; Do not contact and mix the battery module with items that can cause a short circuit, place and store.
- Do not connect the positive and negative terminals of the battery backwards, and it is forbidden to charge the battery system in reverse.
- Be sure to charge or discharge according to the requirements of the product specification, and use charging equipment approved by both parties.
- Ensure that the energy storage converter or charge-discharge equipment used can limit the charging and discharging power according to the allowable power boundary of the battery system, and take corresponding protective measures when the battery charge and discharge reaches the cut-off voltage, and do not overcharge or overdischarge the battery.
- Whether it is charging or discharging, it should be ensured that the battery management system is properly connected and works normally; Ensure the normal communication of the battery management system and avoid abnormal communication of the battery management system due to external line failure or electromagnetic interference.
- When working in a low temperature environment, the capacity and output power capacity of the battery system will be reduced, which is normal; The performance of use gradually recovers after the ambient temperature rises.
- Non-professionals are prohibited from disassembling or modifying the battery management system or battery box to avoid danger.

- The installation and fixation of the battery box and battery cluster should be carried out in accordance with the method of this manual, and the fixing method of the product should not be changed without authorization;
- It is forbidden to mix the battery system with other types of batteries or batteries from other manufacturers.
- If the battery catches fire, it is necessary to use professional firefighting equipment to extinguish the fire, which can be extinguished with dry powder, carbon dioxide, sand, etc. and stay away from the use environment.

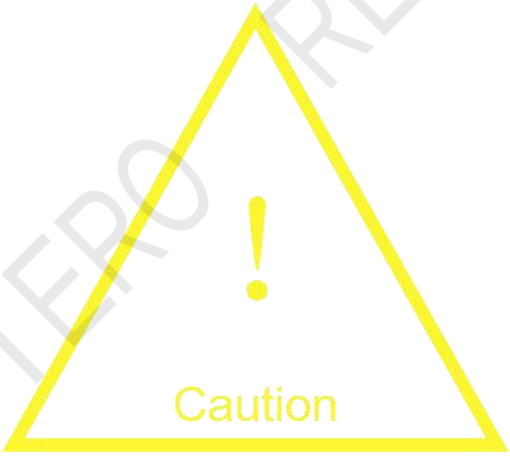
✧ **During installation and maintenance operations, if the manual shows the following signs：**



FAIL TO FOLLOW THE INSTRUCTION WITH THIS SYMBOL MAY RESULT IN A FATAL ACCIDENT, SEVERE INJURY, EVEN DEATH!



FAIL TO FOLLOW THE INSTRUCTION WITH THIS SYMBOL MAY RESULT IN A SERIOUS ACCIDENT OR SEVERE INJURY!



FAIL TO FOLLOW THE INSTRUCTION WITH THIS SYMBOL MAY RESULT IN A MINOR ACCIDENT OR MODERATE INJURY!

✧ Personal protective equipment(PPE)

Including Safety helmet, insulating gloves, protective glasses, protective boots, installation and maintenance clothing：

No.	item	schematic diagram
1	Safety helmet	
2	Insulating gloves	
3	Protective glasses	
4	Protective boots	

5	Installation and maintenance clothing	

Please be aware that a battery can pose a risk of electrical shock including a high short circuit current. Follow all safety precautions while operating the batteries. During the installation or maintenance of the battery system, a worker shall wear proper personal protective equipment such as eye protection, high visibility clothing, protective gloves, and protective footwear. Insulation gloves with over 1500VDC ratings are needed when connecting the busbars and jumpers between modules and racks

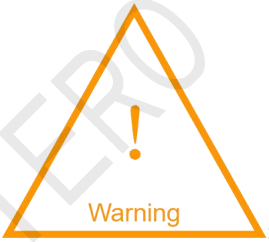
✧ General precaution



：**Please aware that the batteries have a risk of electrical shock with huge short circuit current.**



：**ARC flash hazard is associated with batteries. There is serious risk of ARC flash relating to any equipment modification. Serious injuries can occur in ARC flash incidents.**



:**Please aware that batteries may occur fire in fault condition.**



:Wear personal protective equipment and do not wear any metal object such as watch、rings、metal necklace .etc during performing any operation relating to BESS.



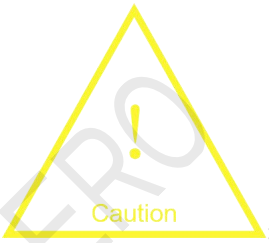
:Use tools with insulated handling to avoid accidental short circuit. Do not put any tool、metal parts on the top cover of battery.



:Do not open or vandalize the battery module



. ： Sharp points and pinch points are present on most system components Be aware of the serious risk of injury while working around equipment batteries.



:Components in the battery system can be damaged by electrostatic discharge. Ensure that you wear a grounded anti-static wrist strap and to discharge static electricity by touching a grounded surface near the equipment before you touch any system components.

III、Disclaimer

Remind customers: REPT BATTERO Energy Co., Ltd.does not assume any free warranty, property damage and safety responsibility for battery performance degradation, life attenuation, damage and safety accidents caused by improper use of lithium-ion batteries in violation of the terms of this article.

- Without the permission of REPT BATTERO Energy Co., Ltd.it is forbidden to apply the battery system in an abnormal environment, including but not limited to the following terms:
  1. Discharge temperature: <-20℃ or >55℃；
  2. Charging temperature: <0℃ or >55℃；
  3. Operating humidity: <10%RH or >90%RH;
  4. Working altitude: >3000m;
  5. Working Environment:

     Areas with high salt spray (e.g. coastal areas);

     Highly contaminated areas （Areas with pollution level 3 and above specified in IEC60664-1).
- Without the permission of REPT BATTERO Energy Co., Ltd ., it is strictly forbidden to disassemble or modify the battery management system or battery box without authorization.
- It is forbidden for the battery system to work in areas close to heat, ignition, and direct sunlight.
- Due to improper operation by on-site personnel (non-our professionals), the battery system suffers mechanical damage, such as deformation, puncture, heavy pressure, fall, etc., REPT BATTERO Energy Co., Ltd . does not assume relevant responsibilities.
- Due to improper operation of on-site personnel (non-our professionals), electrical accidents in the battery system, such as external short circuit, arc pulling, false connection of connectors, damage to parts, etc., REPT BATTERO Energy Co., Ltd .does not bear relevant responsibilities.
- It is forbidden to use the battery beyond the following conditions, including:
  1) Overcharging is prohibited: the upper limit of charging voltage shall not be higher than 3.65V;
  2) Over-discharge is prohibited: the lower limit of discharge voltage shall not be lower than 2.5V;
  3) Overcurrent is prohibited: the continuous, pulse charge and discharge rate shall not exceed the requirements of the charge and discharge current limit table in the specification;
  4) Prohibition of overtemperature: When the battery temperature is lower or higher than the requirements of the charge and discharge current limit table in the specification,

the power should be limited or the charge and discharge should be stopped in time.

- If the customer configures the battery management system by himself, the equipped battery management system should have the monitoring function of battery voltage (monomer), temperature, insulation, current and other parameters, and protect the battery system according to the requirements of the product specification;
- Due to the excessive electromagnetic interference of the working environment of the battery system, the battery system cannot work normally, and REPT BATTERO Energy Co., Ltd .does not assume relevant responsibilities.
- It is forbidden to mix lithium-ion batteries of different types, manufacturers and different chemical systems (such as lead-acid batteries) in the same module, electrical box or other level of system integration products.
- The energy storage battery system integration unit (such as a container) developed by the integrator using the battery box or battery cluster described herein must meet the relevant regulations, standards and specifications of the corresponding application field.
- REPT BATTERO Energy Co., Ltd . does not assume any warranty, property damage and safety accidents caused by battery performance, life attenuation, damage and safety accidents caused by non-battery box or battery cluster factors such as integrated design, thermal design, assembly process, operating environment, abnormal charge and discharge and control.

Ⅳ、Overview of battery system structure



Fig 1. Appearance view

4.1 Main Components parts：

System	Sub Components	Amount	Remark
Bat. rack	20 fts container	1	HQ
	Y104 bat. module	48	BMU inside
	Switch gear	12	SBCU inside
Electrical cabinet	MBCU	1	
	AC & DC power distribution	1	3 phase 380～480Vac±5%
	UPS	1	
	Main switch	1	
Cooling system	Cooling device	1	60kW
	Pipeline system	1	
FSS	Smoke detector	2	
	Gas detector	2	

	Hot Aerosol	96	Module level
	Hot aerosol fire extinguisher	4	Container level
	Fire suppression control panel	1	Container level
	Fire fighting water pipeline	1	Container level

4.2 DC Cabinet

Summary of DC Cabinet

The main functions of the DC cabinet include: output convergence of each battery rack, access and distribution of system auxiliary power supply, uninterruptible power supply, short circuit protection, system data monitoring and display, etc.

Sub Components	Amount	Remark
MBCU	1 pcs	Communication with EMS
Auxiliary AC power distribution	1 set	3 phase 380～480Vac
Auxiliary DC power distribution	1 set	24V
UPS	1 set	
Disconnector	1 pcs	
Fuse	2 pcs	
State lamp	3 pcs	
Battery connection busbar	2 set	

External interface of DC Cabinet

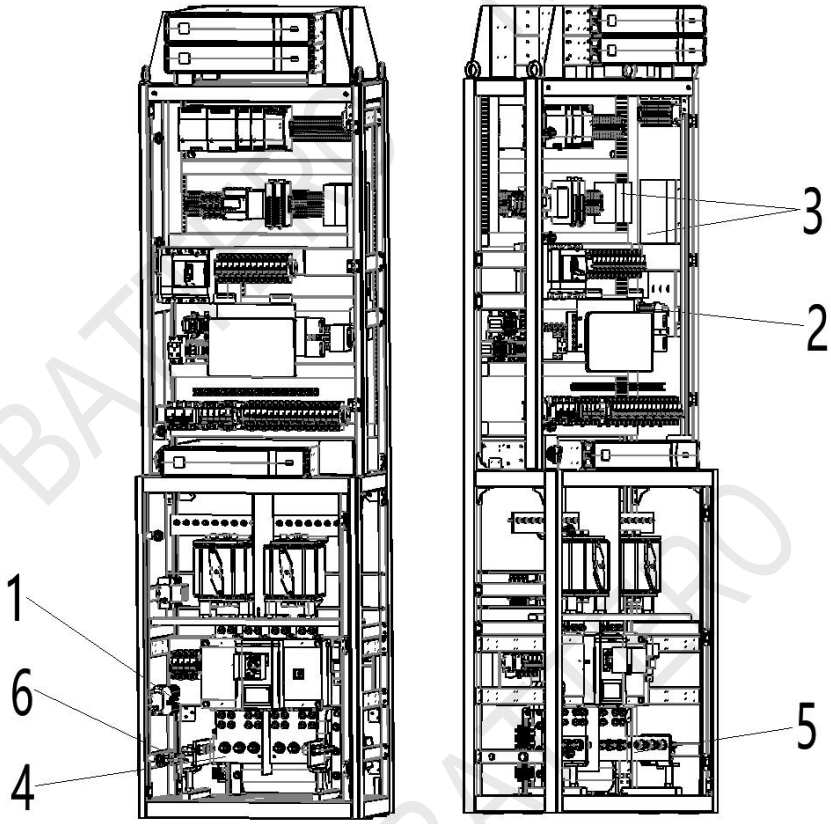


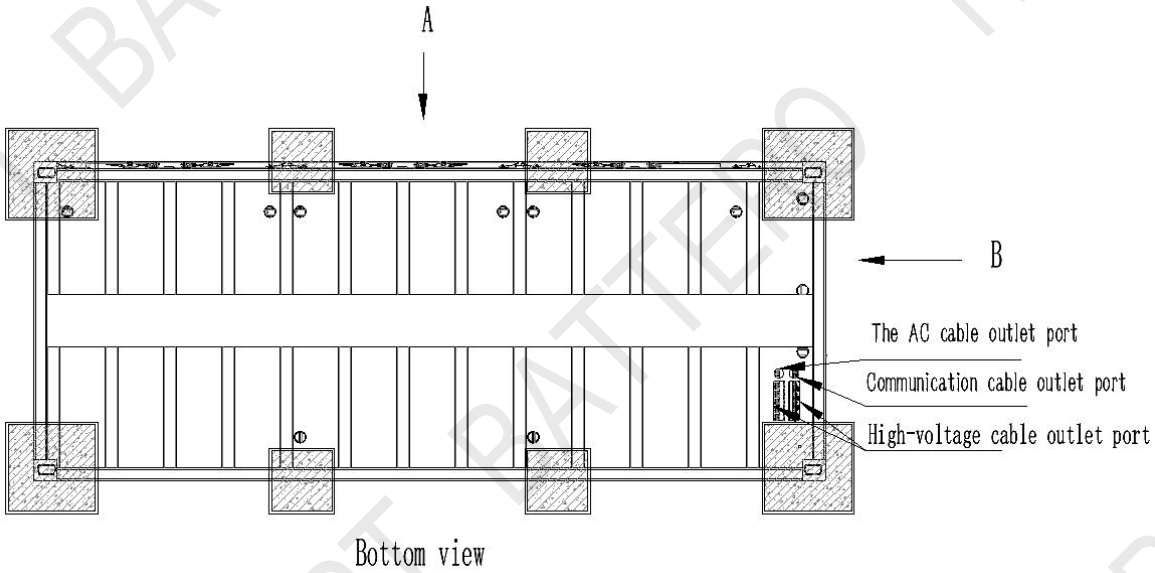
Fig 2 Electric cabinet inside

Electric cabinet wire list

NO.	Function	Terminal definition	Pin function	Hole size	Wire diameter	Remark
1	AC input interface	XT1-1	L1	Ø35mm2	≥25mm2	Bolt crimping
		XT1-2	L2	Ø35mm2	≥25mm2	Bolt crimping
		XT1-3	L3	Ø35mm2	≥25mm2	Bolt crimping
		XT1-4	N	Ø16mm2	≥16mm2	Bolt crimping
		XT1-5	Restart+（optional）	1.5mm2	≥1mm2	Terminal connection 230Vac supply EMS remote restart MBCU
		XT1-6	Restart-（optional）	1.5mm2	≥1mm2	

2	Communication interface(485&CAN&Dry contacts)	XT3-1	Dry contact C	Ø1.5mm2	1mm2	Alarm for normally closed dry contact of PCS1 module machine
		XT3-2	Dry contact C NC	Ø1.5mm2	1mm2	
		XT3-5	RS485.5-A	Ø1.5mm2	1mm2	CAN between BMS and PCS1
		XT3-6	RS485.5-B	Ø1.5mm2	1mm2	
		XT3-7	RS485.5-G	Ø1.5mm2	1mm2	
		XT3-8	GND(earth)	Ø1.5mm2	1mm2	
3	Communication interface(TCP/IP&fiber)	LAN2 or LAN3	1#ES 1#exchange board	CAT6 shield (RJ45)	CAT6 shield (RJ45)	Communication between BMS, EMS （A network)
		LAN2 or LAN3	2#ES 2#exchange board	CAT6 shield (RJ45)	CAT6 shield (RJ45)	Communication between BMS, EMS (B network)
		1#(RX-A)	Fiber box	SMDF（outdoor）	SMDF（outdoor）	Fiber communication between BMS and EMS(A network)
		2#(TX-A)				
		3#(RX-B)		SMDF（outdoor）	SMDF（outdoor）	Fiber communication between BMS and EMS(B network)
		4#(TX-B)				
4	DC+ output bus of 1~12 Racks	P1+	P+	6*M16	240MM2	Configured studs

5	DC- output bus of 1~12 Racks	P1-	P-	6*M16	240MM2	Configured studs"
6	GND bus			12*M6		Configured studs"



Electrical cabinet-PCS DC voltage output:

From the three outlet holes at the bottom of the container (see the figure above for details).

Ⅴ、Battery system storage requirements

- Product storage must be kept away from fire and heat sources. The product is recommended to be adjusted to 40% SOC. When storage time exceeds 2 months, maintenance should be conducted every 2 months.
- If the battery system is not in use for a long time, turn off the system power and disconnect the load. It is recommended to check the energy storage container once every month (1 standard full charge) to ensure that the cell voltage is within 3.2±0.1V range and calibrate the cell SOC. If either cell voltage is lower than 2.8V, it should be charged immediately.
- It is forbidden to connect the positive or negative electrodes of the battery system with metal or other conductors to avoid ignition or short circuit, and mix the battery module with objects that can cause short circuit.

Ⅵ、Battery system transport requirements

- Container assembly transportation shall meet the relevant requirements of UN3536;
- The special lifting point of the container assembly is located at the bottom or the top corner iron, and the load capacity of the lifting equipment meets the requirements;
- Transportation scheme and vehicle selection

Adopt the automobile road transportation scheme, during the execution of the project, go to the project site in advance or through the transportation conditions provided by the demander, select the size of the vehicle according to the road traffic conditions of the project site, and confirm the entry reliability of the primary vehicles with the site project manager of the demander. The battery container transport uses 17.5 meters special vehicle, a battery container. According to the production plan, 7~8 battery containers are transported to the site and transported to the site for consecutive days. The following steps are taken inside the battery module. The loading and assembly scheme is as follows:



1) The battery module is fixed in the battery rack, and the battery module is not easy to become loose.
2) The bottom of the battery frame is fixed to the beam at the bottom of the container, and the top of the battery frame is connected to the beam at the top of the container door frame to ensure that the container does not tilt or collapse during transportation.
3) The DC cabinet, air conditioning, fire cabinet, etc. are connected to the container main beam (load-bearing beam) with the bottom fixed foot, so that they do not tilt or collapse during transportation.
4) Arrange a 17.5-meter trailer to arrange the transportation, and the material weight of each vehicle shall not exceed 60T. This requires part of the battery module before loading, to avoid the hidden danger caused by the bumpy mountain road during the battery transportation.
5) The battery container is reliably limited and fixed on the transport vehicle (there are bound ropes on the vehicle, which can bind the product and the vehicle well) to ensure the safety and reliability during the long-distance transportation process.
6) The whole process of transport vehicles should be stable and average speed, ensure sufficient vehicle spacing, and prohibit sharp brake, sharp turn and other operations, through the speed bump, bumpy road and turning speed is not higher than 15 km/h, to ensure that the container, battery and equipment are not subject to obvious impact and vibration.
7) From the perspective of dangerous goods transportation, the SOC capacity of the battery supplier should not exceed 50% (40% SOC of REPT shipment).

- Transportation safety measures are as follows

Equipment transportation should follow the predetermined route and formulate feasible and safe plans according to different sections, and timely configure command vehicles, large flat vehicles, maintenance work vehicles, police road closure vehicles and logistics support vehicles. In some sections, traffic control should also be implemented on the whole road, and other vehicles should be prohibited from

entering to ensure the smooth passage of the transport fleet. During transportation, warning signs shall be set up for suspension, warning lights shall be set up at night, and special guards shall be set up.

Before bulky transportation, comprehensive and targeted safety education and safety training must be conducted for all participants.

All transportation roads in the construction area shall be equipped with sidewalks and protective railings, separating people and vehicles. Transportation roads should have good drainage facilities and build necessary return sites.

In the early stage of the operation, handle the passage procedures with the local transportation department, and handle the obstacles of traffic facilities before the operation. For aerial cables, they should be equipped with overhead vehicles.

After the fixture of package on vehicle is completed, the transport company shall install and hang the over-limit transport signs and warning signs. The personnel of technology, quality inspection and safety departments and the person in charge of the transportation unit shall thoroughly inspect the products, vehicles, supports and rigging and make written inspection records, and the truck can be sent. After the official shipment, the driver should often get off the bus to check the safety state of the transportation and the protection of the contact parts with the tower body. If there is any abnormality, he should stop immediately.

Before the formal delivery, the person in charge of safety shall disclose the quality and safety technology to the transportation personnel, and clarify the responsibilities and working procedures.

Large truck group should stop for 3-5 hours or 100-200km to inspect. The main inspection items include: vehicle condition inspection and goods bundling inspection, filling in the inspection form. Before loading, the vehicle should be fully inspected and maintained to ensure that state of the vehicle is in the best condition. Check and maintain the tools and rigging used in the transportation to ensure that the performance meets the requirements. In the loading, the operation specifications are strictly implemented. The necessary anti-skid materials between the vehicle and the cargo, the center of gravity of the cargo should be aligned with the bearing center of the vehicle, and the error is within the allowable value range.

The monitoring personnel and the transportation command should pay close attention to the distance between the vehicle, the goods and the obstacles, issue an alarm to the driver in advance, immediately stop the instructions when necessary, until the driver sees the obstacle position and makes clear the correct operation method. In the way of operation, the inspection records of vehicles and goods should be strengthened, and effective measures should be taken to solve the problems found. If the flat car fails, the valve should be closed in time to reduce the height of the vehicle, ensure that the equipment is not damaged, and repair in time. In case of major failure, the parts or vehicle should be replaced in time.

Transportation personnel shall make the following preparations before transportation operation:

(1) Conduct safety education and study the relevant operation procedures and the plan;

(2) Inspect the mechanical equipment and operation tools to ensure the good performance of basic operation facilities;

(3) Provide necessary labor protection articles and inspect;

(4) Confirm the operation tasks.

- Route planning

Follow up the project progress in real time, we will calculate the best plan of the transportation route in advance, choose the shortest route and the best road route, use the expressway or other fast road transportation, assess whether there is overweight, over-limit situation, to ensure that the equipment can arrive at the site smoothly.

- Personnel placement

Before delivering the goods, we shall notify the consignee in advance and provide the contact information of the logistics driver to the project supervisor of the client and provide the real-time report location. Before receiving the goods, if necessary, we shall arrange the after-sales service personnel to go to the site to assist in the acceptance of the goods, and guide the equipment in place, placement, storage and other related work.

Ⅶ、Field installation requirements of the battery system

7.1 Container installation

7.1.1 Foundation requirements

- The recommended foundation height is 300mm away from the floor (set different foundation thickness according to the rainfall in different areas)；
- The recommended number of short concrete columns is 6, the top surface of the base has the same elevation, and the height difference of the container placing surface is 3mm.

7.1.2 Lifting of the energy storage container



：**During the whole lifting process, the container operation safety regulations of the project location country / region must be observed! The container and any machine and tool used in the operation shall be maintained. All personnel engaged in loading and unloading and bolt-fixing should receive corresponding training, especially on safety aspects.**

The following conditions shall be met at the hoisting container site:

- Energy storage container doors are locked;
- Select the appropriate crane or lifting tool according to the site conditions. The selected tool must have sufficient load-bearing capacity and arm length And rotation radius;
- If movement on the slope, etc., additional traction devices may be required;
- Remove all obstacles that exist or may exist in the process of movement, such as trees, cables, etc.;
- The energy storage container should be hoisted under better weather conditions as far as possible;
- Be sure to set up a warning board or warning belt to avoid non-staff members from entering the hoisting area;

7.1.3 Attention



：**In the whole process of lifting the energy storage container, the operation should strictly follow the safety operation procedures of the crane.**



：**No standing person within 5m~10m in the operation area. Especially under the lifting arm and under the lifting or moving machine is strictly prohibited to avoid casualties.**



：**In case of bad weather conditions, such as heavy rain, heavy fog, strong wind, etc., the lifting work should be stopped.**

When lifting the energy storage container, the following requirements:：

- The site safety must be ensured when lifting
- In the lifting and installation operation, the site should have professional personnel to command the whole process;
- The strength of the sling used shall be able to meet the lifting requirements of at least 50t;
- Ensure the safety and reliability of each section connected with the corner;
- The length of the sling can be appropriately adjusted according to the actual requirements of the site;
- The whole lifting process must ensure that the energy storage container is stable, not biased;
- Please use the four top corners of the energy storage container to lift the energy storage container;
- Take all necessary auxiliary measures to ensure the safety and smooth lifting of energy storage containers;

The following figure shows the crane operation diagram during the lifting process of the energy storage container. In the figure, the dotted line circle of the inner layer indicates the operating range of the crane. When the crane works, it is forbidden to stand in the outer solid line circle!

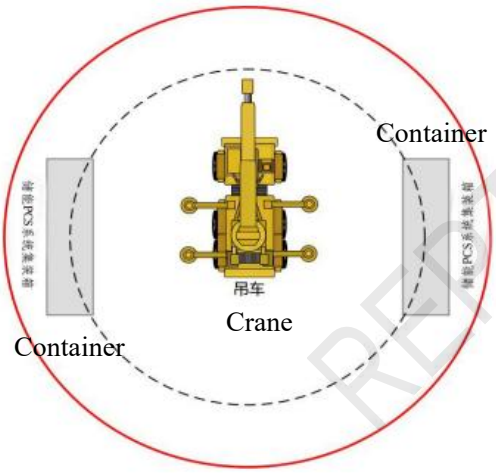


Fig.3 Schematic diagram of the crane operation

7.1.4 Fastening of connectors

Sings with hooks or U-shaped hooks can be used to ceiling the energy storage container. The lifting device shall be connected with the energy storage container box Correct connection of body.

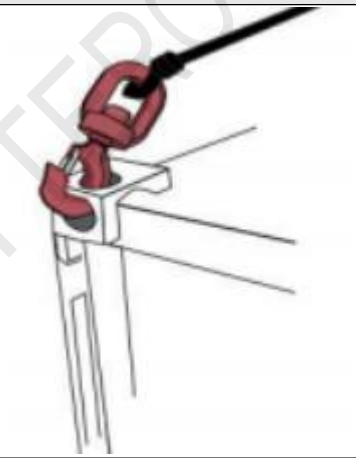
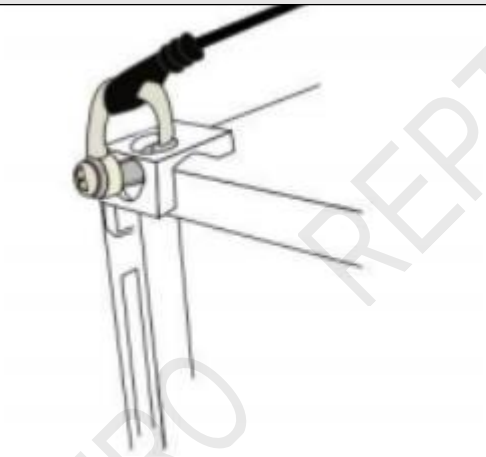
Lift device	Hock	U-shaped hock
Schematic diagram of the connection		
Attention	Should be linked from the inside to the outside, not from the outside to the inside	Horizontal pin must be tightened

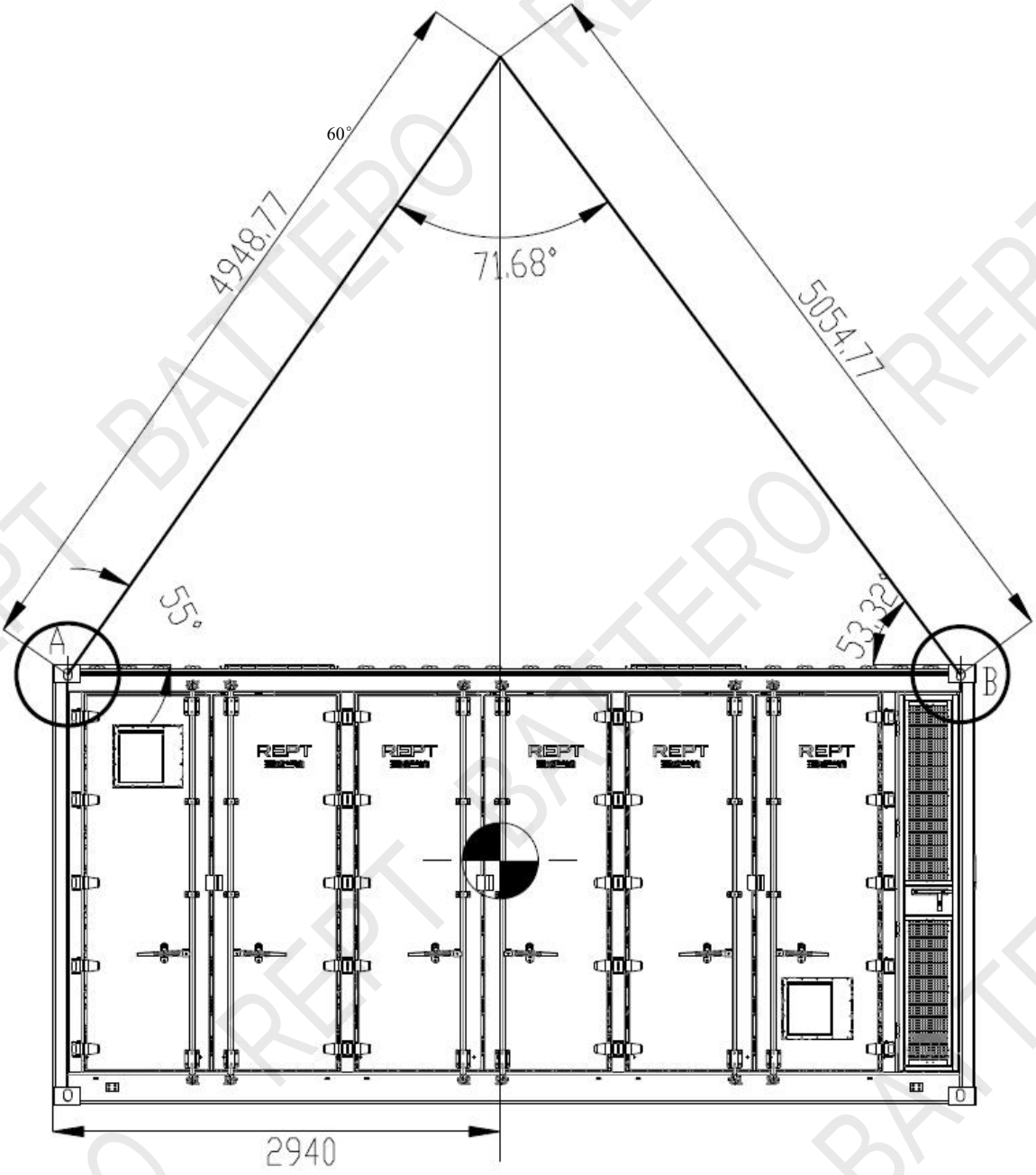
Fig.4 Connection diagram of the lifting device

7.1.5 The lifting process



：<u>When lifting, the safety operation standards and specifications of the project location country / region shall be strictly observed.</u>

7.1.5.1 Top lifting


60°
4948.77
71.68°
5054.77
55°
53.32°
A
B
REPT
REPT
REPT
REPT
REPT
2940

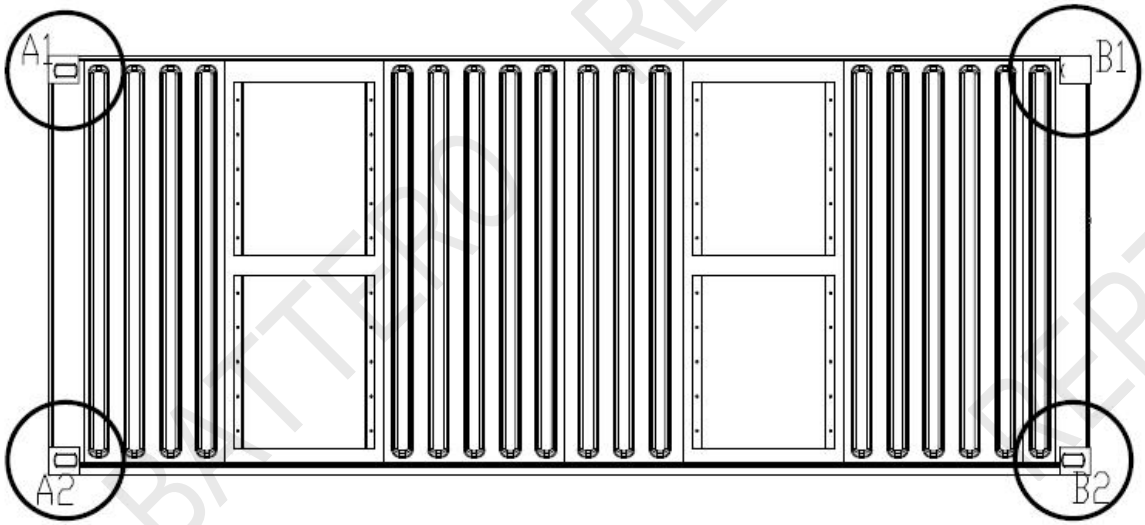


Fig. 5 Top lifting schematic diagram

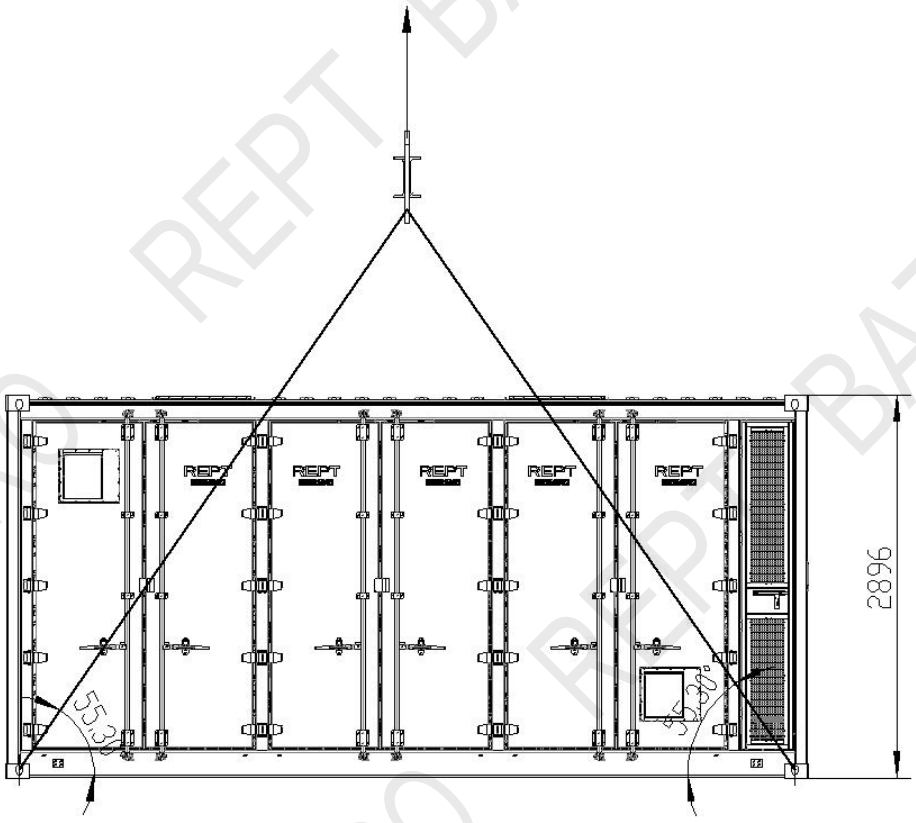
Container hoisting requirements:

Hoisting Angle parts are shown as follows: A (A1,A2),B(B1，B2)

Solution ② Single crane:A point rope 4770mm, B point rope 5380mm

speed 0.5-0.8m/s^2,max1.2m/m^2

7.1.5.2 Bottom lifting



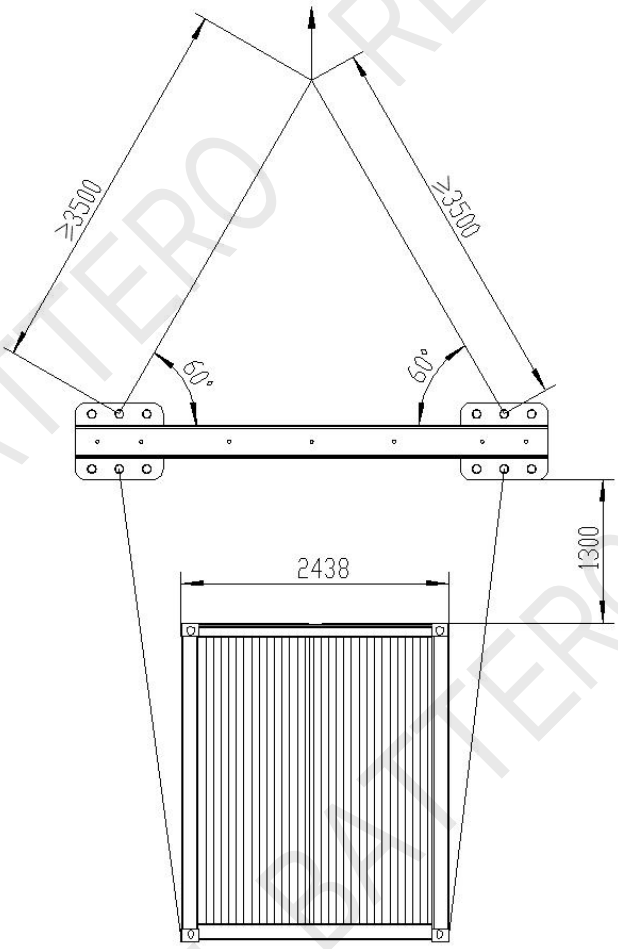


Fig. 6 Bottom lifting schematic diagram

Container hoisting requirements

Hoisting Angle parts are shown as follows:

Solution Single crane:

The lifting direction and angle are shown in the diagram. During lifting, ensure that there are no obstacles around and lift slowly and steadily to ensure safety speed 0.5-0.8m/s^2,max1.2m/s^2.

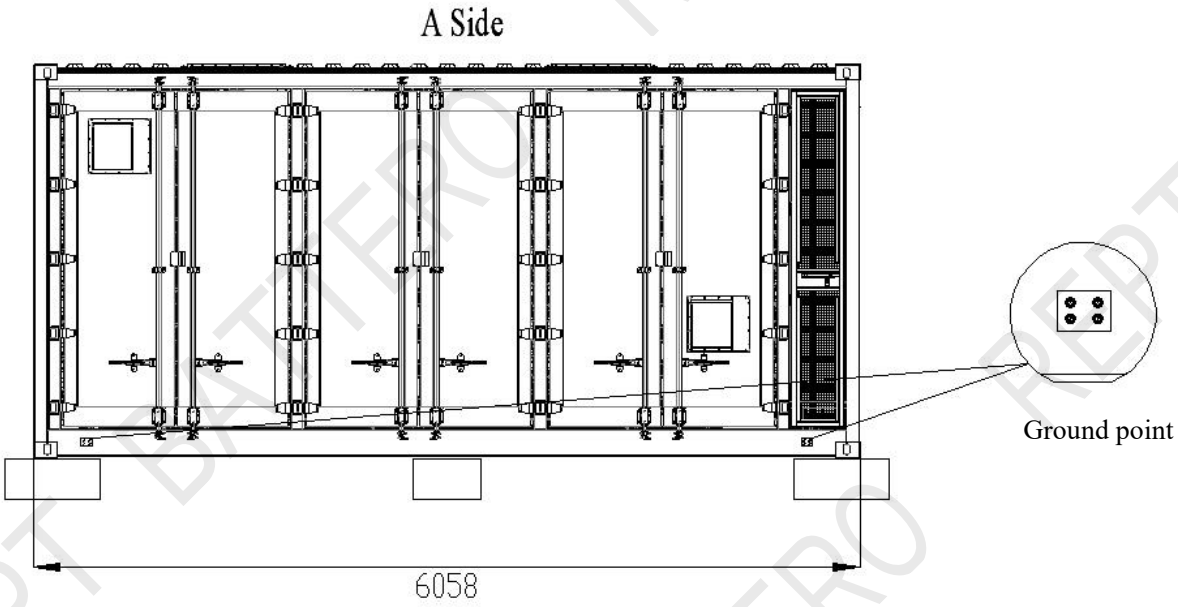
In the process of lifting the energy storage container, each operation link shall be performed according to the following requirements:

- The energy storage container shall be lifted vertically, and shall not be towed on the ground or on the top of the lower box, and shall not be towed in any Push off the energy storage container on the surface;
- After the energy storage container shall be transferred from the support surface of 300mm, and the connection between the spender and the energy storage container shall be suspended Check, after confirming that the connection is firm, before lifting;

- After the energy storage container is in place, it should be placed gently and landed smoothly. It is strictly prohibited to place the container outside the vertical landing by swinging the sling ;
- The site of energy storage container shall be solid and flat, well drained, without obstacles or highlights; energy storage The container shall be supported by only four bottom corners;

7.1.6 Container grounding

- Inside the container provides a main ground conductor system for all metal parts of the equipment main circuit and / or auxiliary circuit, each element is connected by a separate cable. The main grounding system shall be connected with the original grounding system of Party A.
- The container provides two fixing methods of bolt installation and welding. Bolt fixed point and welding point and the non-functional conductive conductor of the whole container (normally uncharged metal shell of the container, etc.), at the same time, the container to the user at least 2 meet the requirements of the most stringent power standards, the user to provide reliable equipotential connection with the non-functional conductive conductor of the whole container, the location is located in the diagonal position of the container.
- The non-functional conductive conductor of the DC junction equipment, its installation base, power distribution cabinet and direct electrical access to the communication monitoring equipment and the installation base shall not be directly connected with the non-functional conductive conductor of the container, but must be uniformly connected with the grounding copper grounding system provided to the container through the container. When the centralized grounding system is disconnected, the casing system and the container
- All contact sites are clearly identified. The roof of the container is equipped with a reliable high-quality lightning protection system. The lightning protection system is connected to no less than two grounding rows provided to the container by the grounding flat steel or the grounding round steel, and the effective cross-sectional area of the conductor in the grounding system is not less than 250mm^2 .
- The ground resistance of all electrical equipment shall be less than 4 ohms.



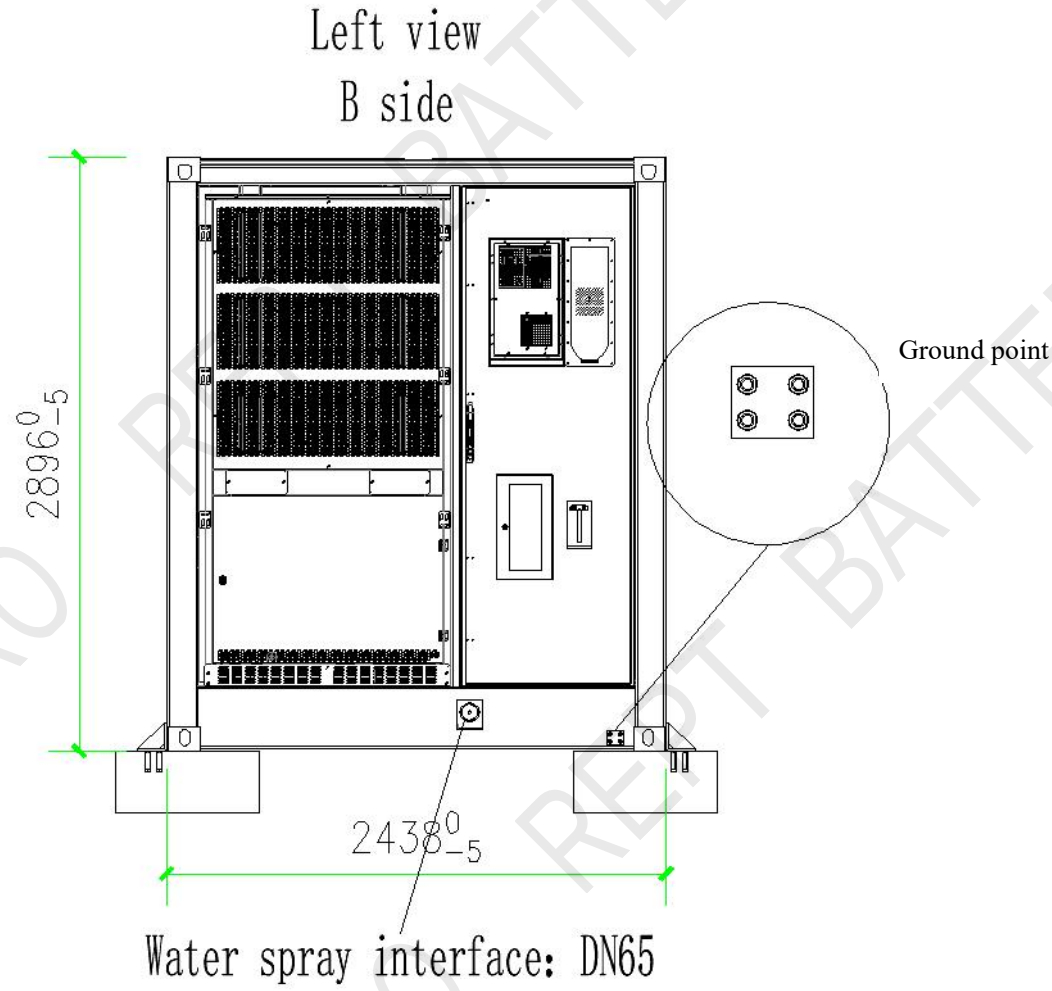


Fig.7 Ground diagram

Connecting location: Grounding location: 4*M12, There are recommended 3 grounding locations for torsion 80 ~ 120 Nm, 2 on A side and 1 on B side, Thread hole depth of 20 mm and select the appropriate bolts according to the thickness of the grounding row; the connection resistance shall be ＜ 0.1Ω.

Ground mode: Adopt a hot-dip galvanized flat steel sheet with a cross-section area of 40mm*4mm, connecting one end to the embedded ground net and the other end to the junction site of the prefabricated module. When the ground grid is embedded, the grounding pad shall be reserved for sufficient length to ensure the connection with the prefabricated module contact site.